



USER MANUAL

Digifiz Replica GEN1 & Digifiz Next User Manual

Product Line Volkswagen Golf® II/III, Vento, Jetta I, II

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PHOL-LABS Kft

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Chapter 1

Precautions

This chapter consolidates the mandatory safety rules that accompany every Digifiz Replica GEN1 and Digifiz Replica Next (or Digifiz Next) instrument cluster. Ignoring any of these items is the fastest way to damage the electronics or obtain unreliable readings.

1. Disconnect the vehicle battery before starting the installation. Working on a powered harness feels faster, but several dashboards have already been destroyed by short circuits caused by a live loom.
2. Never feed the sensor inputs with an external voltage source. The coolant temperature, oil temperature, outside temperature, and fuel level channels are designed for passive sensors only. Even a “harmless” test through a resistor burns the measurement circuitry.
3. Remember that Generation 1 and 1.5 panels have no internal fuse. The first protective element is the 15 A fuse in the Volkswagen fuse box. It reacts far too late to save the cluster from wiring mistakes.
4. Shield the unit from direct sunlight. Prolonged exposure washes out the LCD segments and permanently reduces contrast.
5. Do not attempt to overdrive the LED backlight. Generations 1, 1.5, and 2 use fixed-current lighting. If the daytime image is dim, add shading around the binnacle rather than increasing the drive current.
6. Beware of resonances in cable-driven speedometers. Mechanical drives often oscillate at 40–60 km/h. Fit the supplied electronic sensor—it ships with all current Gen 1.5 and Gen 2 kits—whenever possible.
7. Plan external MFA controls for Generation 2 dashboards. The VW badge touch sensor was removed, so MFA mode switching must come from the steering-column stalk or another external switch.
8. Account for the standby current. A Generation 2 cluster draws roughly 11–13 mA from the vehicle battery even when the ignition is off. This quiescent

consumption cannot be reduced.

9. Instantaneous fuel consumption is not factory fitted. The feature can be retrofitted to Gen 1 and Gen 1.5 units following the instructions below, but it has not been validated for Gen 2 hardware.

<https://www.youtube.com/watch?v=qWqvYc9388U>



(a) Label emphasising battery disconnection during installation.



(b) Warning supplied with the sensor harness against external voltage.

Figure 1.1: Safety placards shipped with the wiring kit.

Chapter 2

Introduction

This operating manual covers the Digifiz Replica GEN1 and Digifiz Replica Next (or Digifiz Next) digital instrument clusters for Volkswagen Golf II/III, Jetta II, Vento, and Scirocco II vehicles. It summarises the hardware variants, describes their functions, and explains how to install, configure, operate, store, and maintain the dashboards. The guidance is intended for vehicle owners, automotive electricians, and service centres that retrofit the product. The expanded edition adds the Golf Mk3 and Volkswagen Vento dashboard configurations that share wiring and calibration procedures with the earlier platforms.

The following chapters provide the product identification scheme, connector pin-outs, operating conditions, and detailed installation and configuration procedures. Maintenance and troubleshooting references for both Replica generations are also included so the instrument panel can be serviced without the factory documentation.

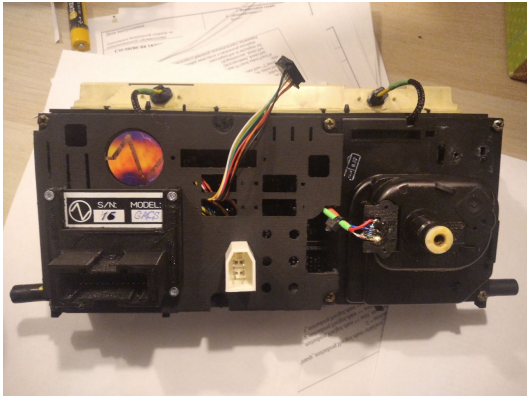
Each variant ships with the components required for the intended drivetrain, measurement units, and wiring harness style. Later chapters decode the variant markings and provide connector tables so the cluster can be integrated safely.



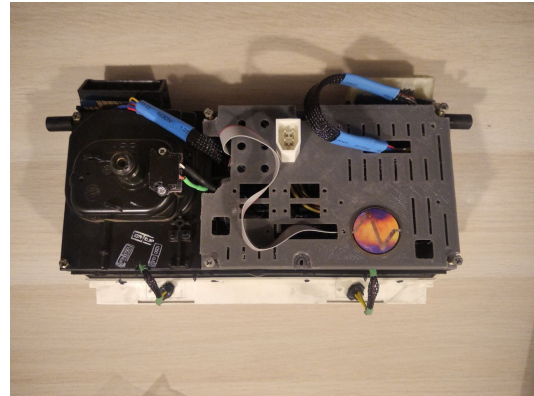
(a) Delivery set for the GART 8-MGF configuration.



(b) Typical contents of a GART package.



(c) Rear view of the single-connector GACS assembly.



(d) Rear view of the dual-connector GACT assembly.

Figure 2.1: Representative Digifiz Replica GEN1 and Digifiz Replica Next (or Digifiz Next) instrument panels supplied with this manual.

Chapter 3

Description and operation of the product

3.1 Purpose

The Digifiz Replica GEN1 and Digifiz Replica Next (or Digifiz Next) dashboards replace the original Volkswagen instrument clusters while extending their functionality. They provide digital indications for speed, engine speed, coolant temperature, fuel level, and auxiliary MFA calculations, and they support both cable and electronic speed sensors. Replica GEN1 units integrate a Bluetooth controller, while Digifiz Next adds Wi-Fi-based configuration modules and optional expansion units.

3.2 Model identification

Every dashboard is marked with a four-letter code that describes the drivetrain, assembly type, speed sensor interface, and wiring generation. Optional digits indicate the supported tachometer scale, and an additional three-letter suffix reports export measurement units.

Four-letter designation

Position 1 G for petrol engines or D for diesel engines.

Position 2 A for factory-assembled units or M for self-assembly kits.

Position 3 C for a mechanical cable speed sensor or R for an electronic speed sensor.

Position 4 T for the pre-facelift (CE 1) harness or S for the facelift (CE 2) harness.

A trailing digit denotes the maximum displayed engine speed in thousands of RPM (for example, “8” on a GACT8 cluster equals an 8000 RPM scale).

Measurement suffix

Export variants may add a three-letter suffix formed from the set MGFK:

M miles per hour,

G gallons,

F Fahrenheit,

K Kelvin.

For example, a GART8-MGF dashboard is a petrol, factory-assembled, electronic-sensor, CE 2 unit with an 8000 RPM tachometer and imperial measurement units.

3.3 Model range

Model	Description
GA CT	Petrol, fully assembled, cable speed sensor, two connectors, 7000 RPM scale.
GA RT	Petrol, fully assembled, remote electronic speed sensor, two connectors, 7000 RPM scale.
GA C	Petrol, fully assembled, cable speed sensor, single connector, 7000 RPM scale.
GA RS	Petrol, fully assembled, remote electronic speed sensor, single connector, 7000 RPM scale.
GA CT8	Petrol, fully assembled, cable speed sensor, two connectors, 8000 RPM scale.
GA RT8	Petrol, fully assembled, remote electronic speed sensor, two connectors, 8000 RPM scale.
GA CS8	Petrol, fully assembled, cable speed sensor, single connector, 8000 RPM scale.
GA RS8	Petrol, fully assembled, remote electronic speed sensor, single connector, 8000 RPM scale.
DA CT	Diesel, fully assembled, cable speed sensor, two connectors, 6000 RPM scale.
DA RT	Diesel, fully assembled, remote electronic speed sensor, two connectors, 6000 RPM scale.
DA CS	Diesel, fully assembled, cable speed sensor, single connector, 6000 RPM scale.
DA RS	Diesel, fully assembled, remote electronic speed sensor, single connector, 6000 RPM scale.
MT	Self-assembly kit with two connectors.
M.S.	Self-assembly kit with a single connector.
NEXT-GA RT	Digifiz Replica Next (or Digifiz Next), 8000 RPM scale, two connectors, electronic speed sensor.
NEXT-GA RS	Digifiz Replica Next (or Digifiz Next), 8000 RPM scale, single connector, electronic speed sensor.
NEXT-MT	Digifiz Replica Next (or Digifiz Next) self-assembly kit with two connectors.
REFIZ	ReFIZ motherboard for OEM Volkswagen Digifiz screen units; replaces the stock motherboard and remains compatible with the Digifiz Next architecture.
NEXT-MS	Digifiz Replica Next (or Digifiz Next) self-assembly kit with a single connector.

3.4 Connector pin-outs

Clusters with two connectors

White connector



Figure 3.1: Connector layout for dual-connector Digifiz Replica GEN1 dashboards.

Pin	Assignment
1	Blinker output, tied to ground for the indicator lamp.
2	Frei — not connected.
3	Terminal 58, positive supply for the panel backlight.
4	Resistive coolant temperature sensor input.
5	Resistive fuel level sensor input.
6	Ground return.
7	Additional ground return.
8	Terminal 1 engine-speed signal (coil, distributor, or other waveform up to 12 V with possible 300 V spikes).
9	MFA mode line used to change MFA functions.
10	UNR permanent positive supply (unused on Replica GEN1, main supply on Digifiz Next).
11	MFA temperature “+” lead for the ambient sensor (Digifiz Next).
12	MFA oil temperature sensor lead (Digifiz Next only).
13	KL 56a high-beam indicator input (+12 V active).

Black connector

Pin	Assignment
1	Terminal 15 switched +12 V from the ignition switch.
2-4	Not connected.
5	Handbrake indicator input (active low).
6	KL 61 generator warning lamp drive with 120 Ω excitation resistor.
7	Oil pressure switch, 0.3 bar.
8	Oil pressure switch, 1.8 bar.
9	Not used.
10	Glow-plug indicator input (+12 V active, diesel only).
11	Hall sensor input for optional speed sensors.
12	MFA block selection line.
13	MFA reset line.

Clusters with a single connector

Single-connector dashboards use the mapping shown in [Figure 3.2](#). The harness replicates the same signals found on the dual-connector variants but consolidates them into a single plug.

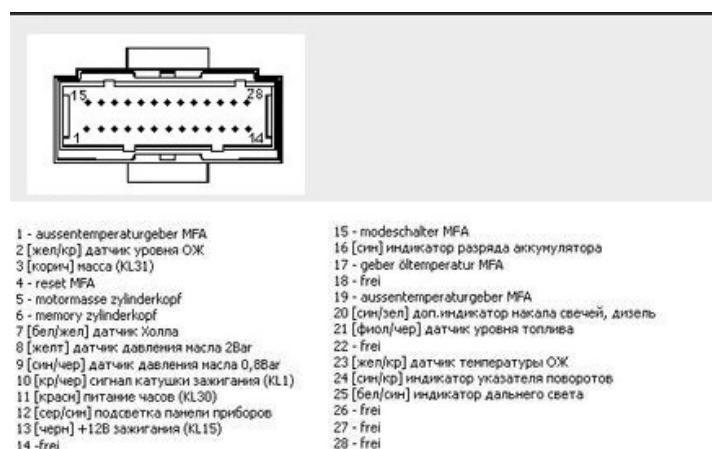


Figure 3.2: Single-connector layout used on compact Replica dashboards.

Scirocco/Passat prospective harness

The prospective Scirocco/Passat harness uses two plugs. Their functions are summarised below.

5-pin plug

Pin	Assignment
1 (D3)	Automatic-transmission “D” range indicator contact. Grounds the drive lamp when the selector is in position D.
2 (D2)	Automatic-transmission second-range indicator contact. Grounds the “2” lamp when the selector is in position 2.
3 (D1)	Automatic-transmission low-range indicator contact. Grounds the “1” lamp when the selector is in position 1.
4 (SA)	Common feed for the automatic selector display (Schaltanzeige); provides the +12 V supply for the range lamps.
5 (SPERRE)	Starter interlock contact from the selector. Closed in park or neutral to permit engine cranking.

14-pin plug

Pin	Assignment
1 (KL 58)	Illumination supply for the panel backlight.
2 (MASS)	Chassis ground return.
3 (TANK)	Fuel-level sender input.
4 (TEMP)	Coolant temperature sender input.
5 (KL 1)	Engine-speed signal (terminal 1).
6 (UHR)	Permanent +12 V feed for the clock and memory backup.
7 (FERNL)	High-beam indicator input.
8 (reserved)	Not connected.
9 (OEL 1.8)	High-pressure oil switch, 1.8 bar.
10 (CAT VORGL(-))	Catalytic-preheat / diesel pre-glow warning lamp input (active low).
11 (OEL 0.3)	Low-pressure oil switch, 0.3 bar.
12 (KL 61)	Alternator warning lamp and excitation feed.
13 (KL 49a)	Combined turn-signal indicator feed.
14 (KL 15)	Ignition-switched +12 V supply.

Mk1 connector mapping

Volkswagen Mk1 vehicles use the following assignments:

1. Illumination and low-beam supply.
2. MASSE 31 ground reference.
3. TANK fuel-level sender.
4. TEMP coolant temperature sender.
5. KL 1 tachometer signal.
6. UHR permanent +12 V.
7. KL 56 high-beam signal.
8. OIL (HIGH) 1.8 bar pressure switch.
9. OIL (LOW) 0.3 bar pressure switch.

- [illegible]

Mk3 Digifiz Next pinout

Pins marked with [†] leave the 28-pin shell through the auxiliary JST breakouts bundled with the Mk3 kit. They provide twisted pairs for the ambient and oil thermistors plus direct access to the MFA buttons and fuel consumption signal.

ReFIZ is a replacement motherboard for original Volkswagen Digifiz screen units. It follows the Digifiz Next architecture, so the web dashboard and all other configuration functions are the same as on Digifiz Next. The stock OEM motherboard can be removed and replaced with ReFIZ. Colour modification is limited to four LED locations plus the indicator line. The fuel-consumption sensor is connected by default, brightness can also be controlled manually through an additional analog

Pin	Assignment
1	Outside air temperature sensor -G17- ground (GND) [†] .
2	Engine Coolant Level (ECL) sensor -G32-.
3	Terminal 31 chassis ground.
4	Multi-function indicator memory switch -E109- reset lead [†] .
5	Terminal 31 chassis ground.
6	Multi-function indicator memory switch -E109- memory lead [†] .
7	Vehicle speed sensor (VSS) output.
8	Oil-pressure switch -F1-, 1.8 bar.
9	Oil-pressure switch -F22-, 0.3 bar.
10	Terminal 1 / terminal W tachometer input.
11	Terminal 30 battery supply (B+).
12	Terminal 58b illumination feed.
13	Terminal 15 ignition-switched B+.
14	Not connected.
15	Multi-function indicator mode-select switch -E86-.
16	Generator warning lamp -K2-, terminal 61.
17	Engine-oil temperature sensor -G8- [†] .
18	Brake / parking-brake warning lamp -K7-.
19	Outside air temperature sensor -G17- signal [†] .
20	Malfunction indicator lamp -K83- or diesel glow-plug lamp -K29-.
21	Fuel gauge -G1-.
22	Left turn-signal indicator lamp -K65-.
23	Engine coolant temperature gauge -G3-.
24	Right turn-signal indicator lamp -K94-.
25	High-beam indicator lamp -K1-.
26	Fuel-consumption input signal [†] .
27	Vehicle speed signal from the speedometer VSS -G22-.
28	Transmission-range (AG4) selector display.

Table 3.4: Pin assignments for the Mk3 Digifiz Next harness.

input, and vehicle signals are routed through the main device socket. An optional CE 2 connector is available as described for the other Digifiz Next harnesses.

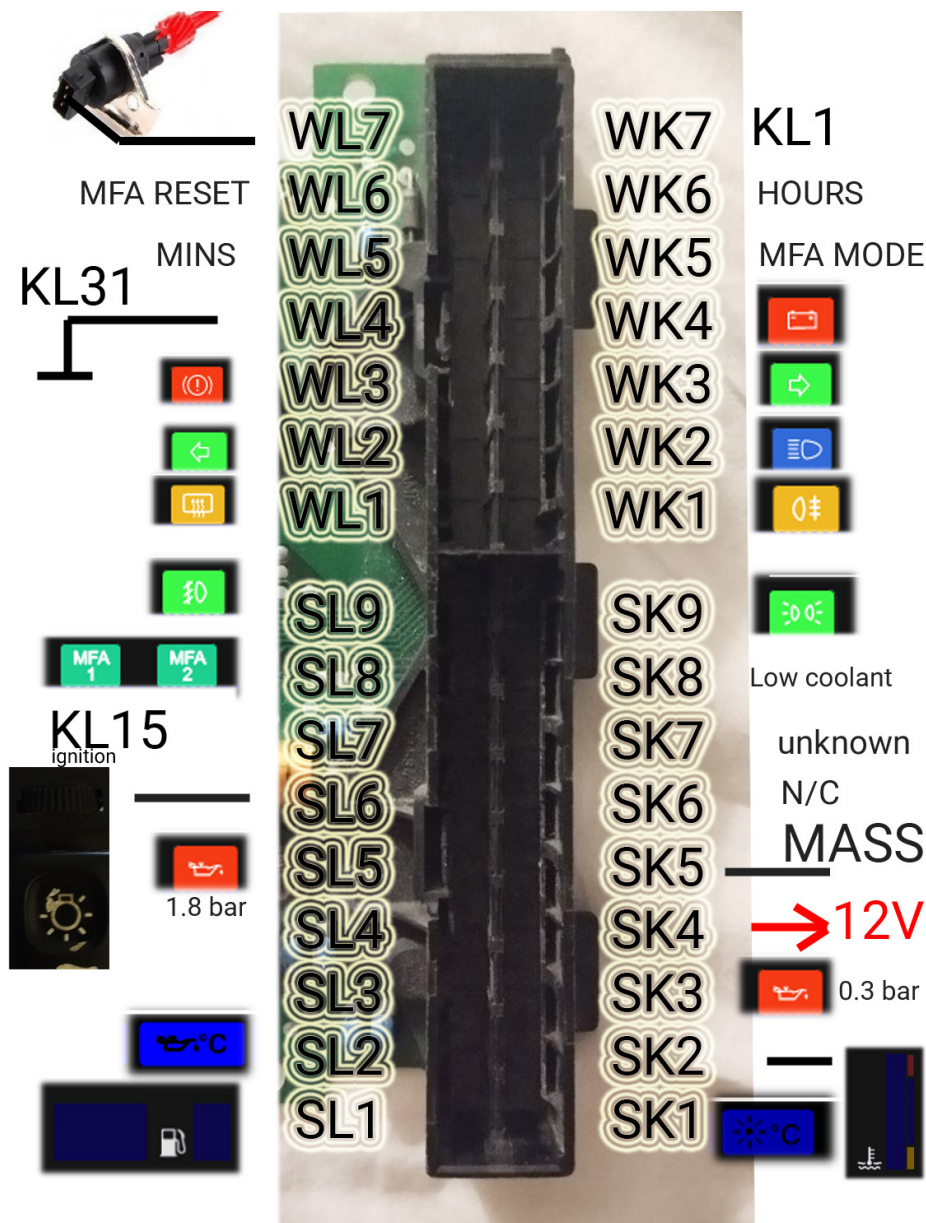


Figure 3.5: ReFIZ motherboard pinout overview.

W Connector – L Side (WL)

Pin	Function	Description
WL7	Speedometer input	Speedometer input from the sensor.
WL6	MFA reset	Resets the Multi-Function Display (MFA) trip computer values.
WL5	Clock – minute setting	Minute adjustment input for the dashboard clock, 12 V level.
WL4	Terminal 31 (Ground)	Chassis ground connection.
WL3	Handbrake warning lamp	Activates parking brake warning indicator.
WL2	Left turn signal indicator	Controls left turn signal lamp in instrument cluster.
WL1	Rear window heater indicator	Indicates rear defroster operation.

W Connector – K Side (WK)

Pin	Function	Description
WK7	KL1 RPM signal	Signal from the coil distributor unit.
WK6	Clock – hour setting	Hour adjustment input for the dashboard clock, 12 V level.
WK5	MFA mode	Switch input for changing MFA display modes.
WK4	Terminal 61	Charging system signal from alternator (battery warning lamp). Generator excitation.
WK3	Right turn signal indicator	Controls right turn signal lamp in instrument cluster.
WK2	High beam indicator	Controls high beam warning lamp.
WK1	Rear fog light indicator	Indicates rear fog lights are switched on.

S Connector – L Side (SL)

Pin	Function	Description
SL9	Terminal 58	Instrument illumination supply from light switch.
SL8	MFA Block switch	Input from the MFA block selection switch (column switch).
SL7	Terminal 15	Ignition-switched +12 V supply.
SL6	Terminal 58b	Dashboard illumination dimmer input.
SL5	Oil pressure switch (1.8 bar)	High-pressure oil switch input.
SL4	N/C; not routed on PCB (non-Rabbit)	Unknown.
SL3	N/C; not routed on PCB (non-Rabbit)	Unknown.
SL2	Oil temperature sender	Sensor input for oil temperature gauge/MFA.
SL1	Fuel level sender	Fuel tank level sensor input.

S Connector – K Side (SK)

Pin	Function	Description
SK9	Terminal 58	Lights indicator.
SK8	Coolant level sensor	Low coolant level signal.
SK7	Unknown +12 V signal input	Unknown; some sources identify it as the variable oil pressure sender output.
SK6	Not connected/unknown	Not routed on the PCB in the non-Rabbit version.
SK5	Terminal 31 (Ground) for MFA	MFA ground connection. Mandatory to avoid malfunction.
SK4	Terminal 30, permanent power	Permanent battery +12 V supply.
SK3	Oil pressure switch (0.3 bar)	Low-pressure oil warning switch input.
SK2	Coolant temperature sender	Engine coolant temperature sensor input.
SK1	Outside temperature sender	Ambient temperature sensor input used by MFA.

The original connectors are difficult to source; a printable socket model is available at:

<https://cults3d.com/en/3d-model/gadget/vw-golf-2-jetta-2-digifiz-sockets-for-the-harness-amp-9265>

Printed-circuit-board service connector

The third connector on the circuit board mirrors the dashboard connectors, with pins numbered right-to-left on Replica GEN1 and Digifiz Next units. It provides a service interface with the assignments listed in [Table 3.6](#).

Position	Assignment
1	Indicator output.
2	Speed sensor input (SPM_M).
3	Vehicle ground.
4	Indicator output.
5	Left blinker optocoupler input.
6	Right blinker optocoupler input.
7	Ignition +12 V.
8	Diesel-specific input.
9	Indicator input (positive).
10	Alternative RPM input (unused, Digifiz Next only).
11	Replica GEN1: indicator output (normally disconnected); Digifiz Next: brake input (active low).
12	Reserved.
13	Check-engine input.
14	No contact.

Table 3.6: Service connector pin assignments.

Suitable electronic speed sensors

Any electronic vehicle speed sensor with a compatible Hall-style output can be used with the dashboard. The recommended sensor is IG0919149A/1H0919149A with mating connector 357972753; when the installation requires the alternate socket style, use 1H0919149C with mating connector 1J0973723.

Compatible alternatives include 191919149D, NTYECP-AU-033, and 357919149B/357919149C commonly used on O2A and O2J gearboxes. These sensors use the same electrical interface and are fully compatible. The colour of the gearbox drive gear is not critical: a sensor supplied with a red gear will still operate correctly even if the gearbox previously used a white gear.

The speedometer signal is connected through the 14-pin (2 x 7) KF2510 additional-signals connector; use the speed-sensor input shown in the connector pinout above.

Auxiliary expansion connectors

Three supplementary four-pin headers are fitted to the main board to simplify harness upgrades and service work:

- Expansion analog signals: provides a dedicated breakout for additional analog inputs when integrating custom sensors.

- MFA mirror: duplicates the standard MFA connector to support parallel tapping of the trip computer signals.
- Analog duplicates: repeats the oil temperature, ambient temperature, and brake indicator inputs so these circuits can be routed to external logging or monitoring modules.

All three use the KF2510-4p mating connector, which is not supplied with the dashboard kit and must be sourced separately if needed.

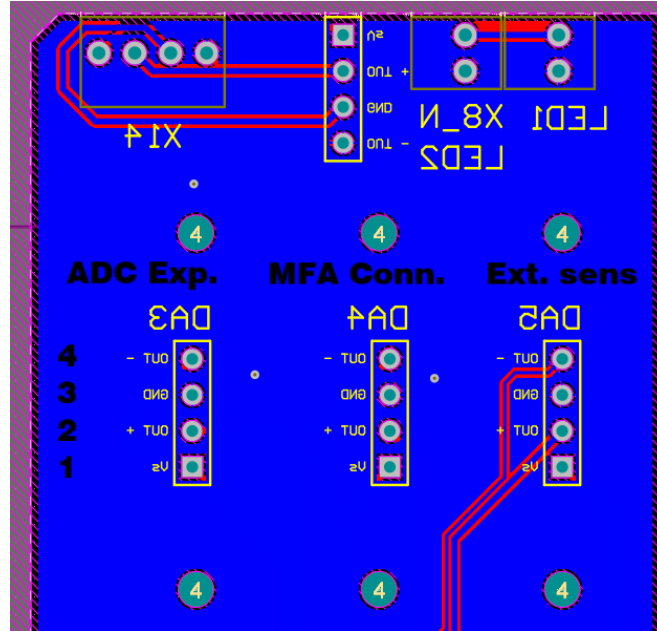


Figure 3.7: Auxiliary connector layout on the main board.

3.5 Embedded software and completeness

The dashboard firmware is published at the following address:

<https://github.com/Sgw32/DigifizReplica>

Two delivery sets are available:

- Digifiz Replica GEN1: dashboard assembly, ambient and oil temperature harness, USBasp programmer, and (for remote sensors) a speed sensor harness.
- Digifiz Replica Next (or Digifiz Next): dashboard assembly and an electronic speed sensor harness.

Connector	Pin	Assignment
Connector I	4	Aux analog input 1
Connector I	3	Ground (GND)
Connector I	2	Aux analog input 2
Connector I	1	VCC (3V3, unfused!!!)
Connector II	4	MFA reset
Connector II	3	Ground (GND)
Connector II	2	MFA memory block
Connector II	1	MFA mode
Connector III	4	Oil temperature sensor output
Connector III	3	Ground (GND)
Connector III	2	Outdoor temperature sensor output
Connector III	1	Brake indicator

Table 3.8: Pin assignments for the auxiliary expansion connectors.

Chapter 4

Operating Principle

Digifiz Replica GEN1 dashboards reuse the original Volkswagen enclosure, the factory CE 1 or CE 2 connectors, and either the mechanical speedometer cable or an electronic speed sensor. Replica GEN1 main boards are based on a fiberglass PCB populated with discrete components controlled by an ATmega 2560 microcontroller and MAX 7219 indicator drivers.

Digifiz Replica Next (or Digifiz Next) builds on an ESP32-S3 system-on-chip and introduces a newly manufactured SLA-printed enclosure, a redesigned front panel and cover, and a connector adapter board. The Digifiz Next display is illuminated by WS2812 addressable LEDs mounted behind the front frame, and the accompanying harness includes the electronic speed sensor by default.

Both generations share the same display layout and MFA pages, ensuring that installation procedures and day-to-day operation remain familiar between hardware revisions.

Chapter 5

Technical specifications

The Digifiz Replica GEN1 dashboard consumes no standby current when powered down. Digifiz Next draws approximately 13 mA from the +12 V supply while the ignition is off, which should be considered when the vehicle is stored for extended periods. Both generations operate reliably from the vehicle electrical system between 9 V and 16 V DC.

5.1 Measurement capabilities

- Vehicle speed: measured through the factory cable or electronic speed sensor. The systematic error is 10 km/h, the relative error is 3 km/h, and the indication saturates at 999 km/h (or mph for imperial units).
- Engine speed: derived from the ignition signal via an optocoupler stage with a 430 nF/1.2 k Ω RC network and a diode limiter. The absolute and relative errors are within 200 rpm.
- Fuel level: read from the resistive tank sender with an uncertainty of approximately 10 litres.
- Coolant temperature: indicated qualitatively using the standard thermistor connected through the vehicle harness; quantitative values are not displayed.
- Timekeeping: maintained to within one minute.
- Indicator lamps: direction indicators, high beam, oil pressure warnings, generator status, handbrake, rear window heating or diesel glow-plug, and front and rear fog lights.

Chapter 6

Operating conditions and safety precautions

6.1 Environmental limits

- The instrument panel operates between -40°C and $+70^{\circ}\text{C}$ at relative humidity up to 95 %.
- The dashboard may remain installed inside the vehicle throughout the year, including when the car is parked for extended periods.

6.2 Safety precautions

1. The Digifiz dashboard is a do-it-yourself device assembled and integrated by enthusiasts. Observe general electrical safety practices while working with it.
2. The product is intended for the personal projects of vehicle owners.
3. The readings are not certified or metrologically verified, although they correspond to the declared specifications at the time of release.
4. Use the dashboard only when you accept responsibility for the installation and for road safety.
5. If the displayed data cannot be trusted, verify it with the vehicle's standard gauges or external measuring instruments.
6. Do not use the instrument panel outputs for automatic vehicle control systems.
7. The authors accept no liability for consequences arising from the installation or use of the dashboard, including traffic fines or accidents. Malfunctions reported within the warranty period (one year for installations performed jointly with the authors and two weeks for independent installations) will be repaired.

8. The functional capabilities listed in Chapter 5^{→ p.21} are guaranteed for one year during supervised installation and for two weeks after independent installation.

Chapter 7

Preparation for work and work order

7.1 Preparing the vehicle

Follow the sequence below when replacing the factory cluster with a Digifiz dashboard:

1. Remove the plastic trim covering the pedals and the lower dashboard to expose the original instrument panel.
2. Disconnect the vehicle battery.
3. Unplug the wiring harness from the factory instrument panel.
4. Detach the mechanical speedometer cable, if present.
5. Unscrew the panel from its brackets and carefully remove it from the vehicle.
6. Route the supplied temperature and speed sensor harnesses as required.
7. Install the Digifiz dashboard into the bracket grooves and secure it with screws.
8. For Digifiz Replica Next (or Digifiz Next), install the Volkswagen MFA sensors (or equivalents) and route their leads to the CE 1/CE 2 connectors.
9. On GACS/GARS/DARS/DACS models, connect the labelled MFA_MODE, MFA_RESET, MFA_BLOCK, and handbrake wires manually if the vehicle harness lacks these contacts. The second-generation Digifiz Next connects these signals internally by default.
10. Plug the harnesses into the dashboard.
11. Fit the electronic speed sensor or reconnect the mechanical cable.
12. Reinstall the dashboard trim and pedal cover in the reverse order.

7.2 Operating the dashboard

- The dashboard powers up automatically with the ignition. The sidelights switch controls the backlight.
- At start-up the entire speed scale illuminates while internal diagnostics stabilise the RPM model; the display then settles on the current idle speed.
- Once the vehicle begins to move, the system reports the parameters listed in [Chapter 5 → p.21](#).

MFA functions

Six MFA pages are available:

1. Daily operating time.
2. Trip distance.
3. Fuel consumption (not implemented on the first Replica revision).
4. Average speed (displayed as the value multiplied by ten).
5. Engine oil temperature (external harness required).
6. Ambient temperature (external harness required).

On Replica GEN1 dashboards a capacitive touch point behind the VW badge cycles the pages; Digifiz Next uses an external steering-column switch. Touch durations behave as follows:

- Short press (< 1 s): cycle to the next MFA function.
- Medium press (1–3 s when no steering-column switch is fitted): switch between MFA memory blocks; the change is indicated on screen.
- Long press (3–7 s): reset the active MFA function (affecting consumption, trip distance, elapsed time, and average speed).

Backlight and indicator layout

The Replica GEN1 dashboard offers a manual brightness trim above the parking-light switch; Digifiz Next relies on automatic brightness driven by a photodiode. Manual overrides can be configured through the maintenance interfaces described in [Chapters 8 and 10 → p.??](#).

The layout of the horizontal indicator block and the on-screen legend are shown in [Figure 7.1](#).

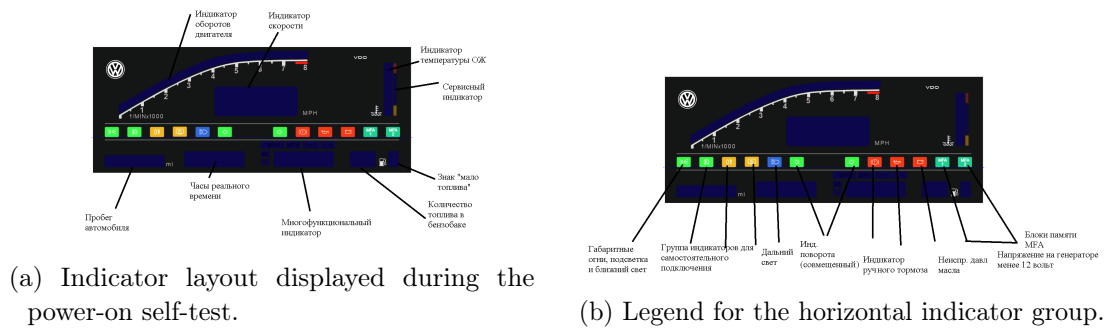


Figure 7.1: Instrument panel indication scheme.

Configuration interfaces

- Classic Digifiz Replica GEN1 units include a Bluetooth 2.0 (or BLE-compatible) module. Install the Serial Bluetooth Terminal application from Google Play, pair with the dashboard, and issue commands directly from the terminal view. Apple iOS devices cannot connect to this module.
- Digifiz Next exposes an embedded Wi-Fi access point and configuration portal described in [Chapter 8 → p.27](#). Disable mobile data while connecting to ensure the captive portal loads correctly.

Both generations can also be powered and configured on the bench using the USBasp programming interface.

Chapter 8

Setting up and maintaining Digifiz Replica Next (or Digifiz Next)

This section applies to the Digifiz Replica Next (or Digifiz Next) dashboard shown in [Figure 8.1](#).



Figure 8.1: Digifiz Replica Next (or Digifiz Next) dashboard assembly.

8.1 Panel handling

- The UV-printed polycarbonate faceplate must be protected from scratches and foreign objects. Significant damage requires replacement parts from PHOL-LABS Kft and is not treated as a warranty case.
- The real-time clock is configured via the Wi-Fi control panel. It resets whenever the permanent supply is disconnected.

8.2 Wi-Fi control portal

Configuration, data collection, and firmware management are performed through the embedded web application.

- Connect to the dashboard's Wi-Fi access point. Disable mobile data and join Digifiz_AP (password 87654321); some revisions advertise PHOL-LABS2 with the same password.
- The default IP address is 192.168.4.1. If the dashboard is configured to join another network, scan the subnet for an address ending in .32 using an IP tools application.
- The portal contains five tabs: WiFi, Control, Settings, Colors, and About (Figure 8.2). The Wi-Fi tab configures network settings and handles firmware uploads; the Control tab adjusts dashboard parameters; the Settings tab provides a structured editor for all firmware parameters; the Colors tab manages multi-segment colour schemes; the About tab lists author information.

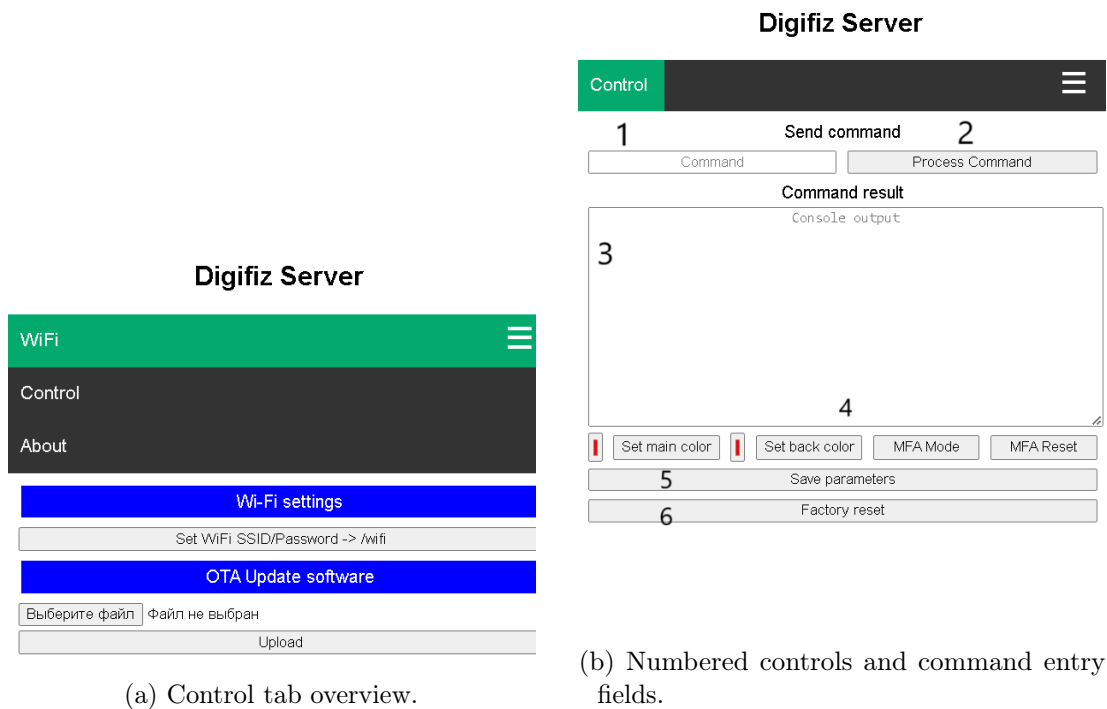


Figure 8.2: Digifiz Next Wi-Fi control interface.

8.3 Command entry

The Control tab provides a command input line (1), a Process button (2), a result window (3), quick controls (4), a Save button (5), and a Reset button (6). Enter commands as space-separated pairs <number> <value> using integers only; punctuation and quotation marks are not required. Figure 8.3 illustrates the interface while toggling automatic brightness.

Command	Name	Description
22 (or 0)	PARAMETER_RPMCOEFFICIENT	Engine RPM calibration factor (100–10000).
1	PARAMETER_SPEEDCOEFFICIENT	Speed calibration factor (10–255).
2	PARAMETER_COOLANTTHERMISTORB	Coolant thermistor beta coefficient (2000–5000).
3	PARAMETER_OILTHERMISTORB	Oil thermistor beta coefficient (2000–5000).
4	PARAMETER_AIRTHERMISTORB	Ambient thermistor beta coefficient (2000–5000).
5	PARAMETER_TANKMINRESISTANCE	Minimum fuel sender resistance (0–1000 Ω).
6	PARAMETER_TANKMAXRESISTANCE	Maximum fuel sender resistance (100–1000 Ω).
7	PARAMETER_TAU_COOLANT	Coolant temperature filter constant (1–50, higher is more responsive).
8	PARAMETER_TAU_OIL	Oil temperature filter constant (1–50).
9	PARAMETER_TAU_AIR	Ambient temperature filter constant (1–50).
10	PARAMETER_TAU_TANK	Fuel level filter constant (1–50).
11	PARAMETER_MILEAGE	Total odometer value (0–999999).
12	PARAMETER_DAILY_MILEAGE	Trip odometer (0–9999).
13	PARAMETER_AUTO_BRIGHTNESS	Automatic brightness enable (1=on, 0=off).
14	PARAMETER_BRIGHTNESS_LEVEL	Manual brightness level (0–60%; values above 60 reduce LED life).
15	PARAMETER_TANK_CAPACITY	Fuel tank capacity in litres (0–99; 55 L typical for Golf II/III and Vento).
16	PARAMETER_MFA_STATE	Active MFA mode (normally controlled via hardware input).
17	PARAMETER_BUZZER_OFF	Disable buzzer (1 disables, 0 enables; Digifiz Next lacks a buzzer).
18	PARAMETER_MAX_RPM	Tachometer scaling (typical 8000, range 4000–16000).
19	PARAMETER_NORMAL_RESISTANCE_COOLANT	Coolant sensor resistance at 25 °C (1000–10000 Ω).
20	PARAMETER_NORMAL_RESISTANCE_OIL	Oil sensor resistance at 25 °C (1000–10000 Ω).
21	PARAMETER_NORMAL_RESISTANCE_AMB	Ambient sensor resistance at 25 °C (1000–10000 Ω).
23	PARAMETER_DOT_OFF	Clock colon behaviour (0=blink, 1=solid).
24	PARAMETER_BACKLIGHT_ON	Enable backlight on low beam (not used on Digifiz Next).
25	PARAMETER_M_D_FILTER	Median filter constant (legacy, normally unused).
26	PARAMETER_COOLANT_MAX_R	Coolant sensor threshold for full-scale indication (100 °C–150 °C).
27	PARAMETER_COOLANT_MIN_R	Coolant sensor threshold for “1 bar” indication (0 °C–80 °C).
31	PARAMETER_MAINCOLOR_R	Red component of the UI colour (0–255).
32	PARAMETER_MAINCOLOR_G	Green component of the UI colour (0–255).
33	PARAMETER_MAINCOLOR_B	Blue component of the UI colour (0–255).
37	PARAMETER_RPM_FILTER	RPM filter aggressiveness (10–200, higher reacts faster).
128	PARAMETER_READ_ADDITION	Add 128 to read the current value of any command.
255	PARAMETER_SET_HOUR	Set clock hours (24-hour format).
254	PARAMETER_SET_MINUTE	Set clock minutes.
253	PARAMETER_RESET_DAILY_MILEAGE	Reset the trip odometer.
252	PARAMETER_RESET_DIGITAL	Factory reset of stored parameters.

Table 8.4: Primary Digifiz Next configuration commands.

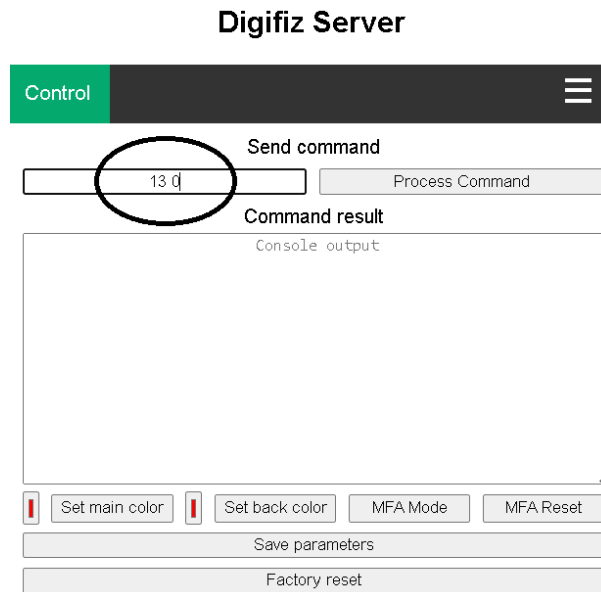


Figure 8.3: Example command sequence disabling automatic brightness.

8.4 Command reference

8.5 Default values

8.6 Reading parameters and examples

To read a parameter, add 128 to the command number (for example, 129 0 reports the speed coefficient). Typical commands include disabling automatic brightness (13 0), enabling it again (13 1), adjusting the speed coefficient (1 110 increases the displayed speed by 10%), and setting the odometer (11 123456). Clock values are set with 255 <hours> followed by 254 <minutes>. Commands 31–33 set the RGB components of the user interface colour.

8.7 Service commands

Recent firmware revisions accept human-readable parameter names, for example `PARAMETER_RPMCOEFFICIENT 3000`. The diagnostic command `adc 0` prints raw ADC readings for sensor troubleshooting. Firmware updates add visual colour controls, so update regularly through the WiFi tab to access the latest features.

Parameter	Default	Notes
PARAMETER_RPMCOEFFICIENT	3000	Typical for Audi tachometer inputs.
PARAMETER_SPEEDCOEFFICIENT	100	Calibrated for 100 km/h.
PARAMETER_COOLANTTHERMISTORB	4000	
PARAMETER_OILTHERMISTORB	4000	
PARAMETER_AIRTHERMISTORB	3812	3600 for Generation 2 panels.
PARAMETER_TANKMINRESISTANCE	35	Ω .
PARAMETER_TANKMAXRESISTANCE	265	Ω .
PARAMETER_TAU_COOLANT	2	Filter constant.
PARAMETER_TAU_OIL	2	Filter constant.
PARAMETER_TAU_AIR	2	Filter constant.
PARAMETER_TAU_TANK	2	Filter constant.
PARAMETER_MILEAGE	Vehicle-specific	Retains stored odometer.
PARAMETER_DAILY_MILEAGE	0	
PARAMETER_AUTO_BRIGHTNESS	1	Enabled.
PARAMETER_BRIGHTNESS_LEVEL	25	Generation 2 default; Generation 1/1.5 use 7 or 13.
PARAMETER_TANK_CAPACITY	63	Litres.
PARAMETER_MFA_STATE	0	Default MFA page.
PARAMETER_BUZZER_OFF	1	Buzzer disabled.
PARAMETER_MAX_RPM	8000	Tachometer scale.
PARAMETER_NORMAL_RESISTANCE_COOLANT	1000	Ω at 25 °C.
PARAMETER_NORMAL_RESISTANCE_OIL	1000	Ω at 25 °C.
PARAMETER_NORMAL_RESISTANCE_AMB	2991	500 Ω for Generation 2 sensors.
PARAMETER_DOT_OFF	0	Blinking clock colon.
PARAMETER_BACKLIGHT_ON	1	Backlight enabled with low beam.
PARAMETER_M_D_FILTER	65535	Legacy median filter constant.
PARAMETER_COOLANT_MAX_R	120	°C.
PARAMETER_COOLANT_MIN_R	60	°C.
PARAMETER_MAINCOLOR_R	180	Yellow-green default.
PARAMETER_MAINCOLOR_G	240	Yellow-green default.
PARAMETER_MAINCOLOR_B	6	Yellow-green default.
PARAMETER_RPM_FILTER	70	Filter response.
PARAMETER_UPTIME	0	Runtime counter.

Table 8.5: Digifiz Next default settings.

8.8 Settings tab parameter editor

The Settings tab mirrors the parameter list in [Table 8.4](#) and [Table 8.5](#) while adding metadata about ranges, descriptions, and data types. [Table B.1](#) summarises every firmware parameter. Use the Settings tab whenever you prefer a graphical workflow instead of typing command numbers; it exposes the same catalogue, including range validation, so you can adjust values quickly without memorising numeric IDs.

1. Press Load Parameters to fetch the live values from the dashboard. The browser displays each item with its name, current value, tooltip, and type.
2. For numeric entries, type the desired value inside the New Value column. The interface enforces the permissible range shown in the Min and Max columns. Boolean parameters appear as checkboxes.
3. Click Set to submit the change instantly. The table refreshes to confirm the updated value.
4. Repeat the process for every parameter you want to adjust. When finished, return to the Control tab and press Save parameters to write the configuration to non-volatile memory.

The colour workflow requires that you enable the firmware flag responsible for custom palettes before moving to the Colors tab. Locate the boolean entry labelled “Custom colour scheme” in the web dashboard’s Settings tab, tick the checkbox, and press Set. The dashboard will reject bespoke segment edits until this flag is turned on.

8.9 Custom colour schemes

The Colors tab introduces a segment-based editor for the WS2812 LED backlight. Each row describes a range end point, the functional area it corresponds to, and the colour type used for that range. In the colour type selector, Main color uses the RGB value currently set on the Control tab, Main color /3 uses the same main colour at one third of its intensity (three times darker), Custom color uses the RGB value selected directly on this Colors tab, and Back color uses the backlight colour from the Control tab.

1. Press Load Scheme to read the active mapping. Use Add Segment, or the inline “+↑” and “+↓” controls, to insert new ranges. The drop-down menus select the instrument function, and the base-colour selector lets you reuse the main or back colours instead of specifying a fixed RGB value.
2. Click the colour picker to fine-tune the RGB tone for segments that are set to “Custom”. The editor reports component values in real time.
3. Use the reorder arrows to match the physical LED sequence (segments must

remain in ascending order). Delete redundant rows with the “×” button.

4. When the table reflects the desired layout, press Set Scheme. The browser iterates through the rows and pushes each segment to the dashboard.
5. Immediately switch to the Control tab and press Save parameters. This step is mandatory—the firmware caches the uploaded segments in RAM and discards them after a reboot if they are not saved.
6. Optionally export the JSON representation via Export to File for backups, or import a previously saved file with Import from File. The Reset Scheme button restores the factory layout after confirmation.

If you later disable the custom colour scheme flag in the Settings tab, the dashboard falls back to the classic single-colour mode driven by `PARAMETER_MAINCOLOR_R/G/B`■

Chapter 9

Typical situations for setting up the Digifiz Next

Hotspot not visible Move closer to the vehicle and ensure it is parked in an open area. Disable mobile data, forget stale Wi-Fi profiles, and reconnect to Digifiz_AP (or PHOL-LABS2).

404 at 192.168.4.1 Turn off mobile data on the phone or laptop and reload the page. Captive portal detection on Android/iOS often interferes until the cellular modem is disabled.

Firmware updates Open the WiFi tab and select the supplied Digifiz.bin file. The latest releases are published at the link below.

<https://github.com/Sgw32/DigifizReplica/releases>

Click Upload. The first attempt can fail; repeat the upload if necessary. Successful flashes redirect to a confirmation page. Record the odometer before updating and restore it afterwards with 11 <mileage>.

Commands ignored Refresh the browser, return to the Control tab, and resend the command. Ensure the Process button is pressed after entering the value.

Speed reading incorrect Connect via Wi-Fi, drive at an indicated 100 km/h, note the GPS speed, then issue 1 <gps_value> (for example, 1 85) to set `PARAMETER_SPEEDCOEFFICIENT` to the verified value.

RPM reading incorrect Adjust `PARAMETER_RPMCOEFFICIENT`. Older firmware uses 0 <value>; current versions use 22 <value>. Example: 22 1500 halves the reading relative to 22 3000.

Display too dim Disable automatic brightness with 13 0, then raise the manual level (for example, 14 50). Experiment with values between 45 and 55; avoid levels above 60 to preserve LED life.

Setting the clock Use the web terminal (or Serial Bluetooth Terminal on legacy

builds) to send 255 <hours> followed by 254 <minutes>. Example: 255 23 and 254 55 sets 23:55.

Fuel readings stuck Disconnect the battery and measure the resistance between the fuel sender pin and vehicle ground. Valid readings are typically 30–300 Ω . Repair shorts below 5 Ω or open circuits before reconnecting. If the readings vary correctly but the gauge does not, record adc 0 results at several fuel levels and share them with PHOL-LABS Kft.

Fuel flow readings inaccurate The optional flow sensor produces emulated data and is unreliable without an intake manifold pressure sensor. Treat the readings as experimental.

Coolant temperature out of range Tune `PARAMETER_COOLANT_MIN_R` and `PARAMETER_COOLANT_MAX_R`. Example: 27 30 lowers the “1 bar” threshold to 30 °C.

Oil or ambient temperature missing With the battery disconnected and the engine cold, measure the sensor resistance. Oil sensors should read about 2 k Ω $\pm$ 0.3 k Ω , ambient sensors about 10 k Ω $\pm$ 2 k Ω . Adjust `PARAMETER_NORMAL_RESISTANCE_OIL` (command 20) or `PARAMETER_NORMAL_RESISTANCE_AMB` (command 21); lower values decrease the indicated temperature, higher values increase it. Persistent issues should be diagnosed by collecting adc 0 output and contacting PHOL-LABS Kft.

Changing interface colour Use commands 31–33 to set the RGB values. New firmware revisions include visual colour controls in the web interface, so update regularly.

Chapter 10

Setup and maintenance of the Digifiz Replica GEN1

This chapter applies to the classic Digifiz Replica GEN1 instrument panel shown in [Figure 10.1](#). If your dashboard matches the Digifiz Replica Next (or Digifiz Next) layout, refer to the previous chapter.



(a) Classic Digifiz Replica GEN1 with square bezel.



(b) Round-edge fascia used on later kits.

Figure 10.1: Appearance of the Digifiz Replica GEN1 dashboard.

10.1 Handling and screen care

- The plexiglass front with UV printing is easily marred. Avoid contact with sharp or abrasive objects.
- Surface damage is cosmetic and not covered by warranty. Request replacement parts from PHOL-LABS Kft if the screen pattern is deformed.

10.2 Real-time clock battery

The dashboard contains a DS3231 real-time clock with a CR2032 cell. The battery typically lasts about four years. When it is depleted the clock resets at every power-up. Remove the front and/or rear cover, keep the wiring harnesses connected, and replace the coin cell. Dispose of the spent battery according to local regulations.

10.3 Firmware maintenance with USBasp

Each kit ships with a USBasp programmer lead already connected inside the housing (Figure 10.2). Install a suitable USBasp driver before flashing. For example, download it from the following address:

<https://myrobot.ru/downloads/driver-usbasp-v-2.0-usb-isp-windows-7-8-10-xp.php>

The programmer powers the dashboard when it is connected to a computer, allowing bench checks.

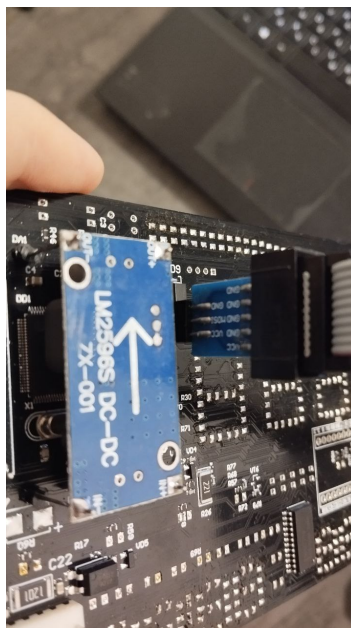


Figure 10.2: USBasp harness orientation inside the Digifiz Replica GEN1.

Flash firmware with avrdude using the command below (replace the firmware filename if required):

```
avrdude -c usbasp -p m2560 -e \  
-U lfuse:w:0xff:m -U hfuse:w:0x99:m -U efuse:w:0xff:m \  
-U flash:w:Digifiz.ino.mega.hex
```

After a successful upload press the front touch button four to five times to initialise the memory blocks. If the blocks remain empty, repeat the flashing procedure or issue the Bluetooth command 252 0 to trigger a factory reset. Ready-to-use firmware images are published at:

<https://github.com/Sgw32/DigifizReplica>

10.4 Bluetooth configuration

Most parameters are adjusted over Bluetooth using an Android phone and the Serial Bluetooth Terminal application. Download it from the following link before pairing with the dashboard:

https://play.google.com/store/apps/details?id=de.kai_morich.serial_bluetooth_terminal&hl=en&gl=US

iOS devices cannot connect to the classic Bluetooth 2.0 module.

- Ensure you pair with the dashboard's Bluetooth Classic interface rather than BLE-only devices.
- In Serial Bluetooth Terminal set the end-of-line character to LF. Disable CR+LF before sending commands.

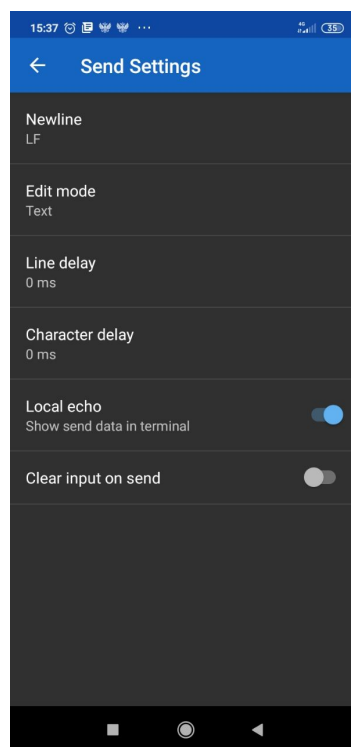


Figure 10.3: Recommended Serial Bluetooth Terminal configuration.

Enter commands as space-separated pairs `<number> <value>`. For example, to store an odometer value of 123 456 km send 11 123456. Add 128 to a command number to read its current value (129 0 reports the speed coefficient). The diagnostic command `adc 0` prints raw sensor readings that help the developers analyse faults.

10.5 Configuration parameters

The primary Bluetooth commands are listed in [Table 10.4](#). Default settings for Generation 1/1.5 and Generation 2 dashboards are summarised in [Table A.2](#). Use commands 31–33 only on Digifiz Next units; they have no effect on the classic Replica GEN1.

Table 10.4: Classic Digifiz Replica GEN1 configuration commands.

ID	Name	Description
22 (or 0)	PARAMETER_RPMCOEFFICIENT	Engine RPM calibration factor.
1	PARAMETER_SPEEDCOEFFICIENT	Speed calibration factor.
2	PARAMETER_COOLANTTHERMISTORB	Coolant thermistor beta coefficient.
3	PARAMETER_OILTHERMISTORB	Oil thermistor beta coefficient.
4	PARAMETER_AIRTHERMISTORB	Ambient thermistor beta coefficient.
5	PARAMETER_TANKMINRESISTANCE	Minimum fuel sender resistance (Ω).
6	PARAMETER_TANKMAXRESISTANCE	Maximum fuel sender resistance (Ω).
7	PARAMETER_TAU_COOLANT	Coolant temperature filter constant.
8	PARAMETER_TAU_OIL	Oil temperature filter constant.
9	PARAMETER_TAU_AIR	Ambient temperature filter constant.
10	PARAMETER_TAU_TANK	Fuel level filter constant.
11	PARAMETER_MILEAGE	Total odometer value.
12	PARAMETER_DAILY_MILEAGE	Trip odometer.
13	PARAMETER_AUTO_BRIGHTNESS	Enable automatic brightness adjustment.
14	PARAMETER_BRIGHTNESS_LEVEL	Manual brightness level (0–15).
15	PARAMETER_TANK_CAPACITY	Fuel tank capacity (litres).
16	PARAMETER_MFA_STATE	Active MFA page.
17	PARAMETER_BUZZER_OFF	Disable the buzzer (1 disables, 0 enables).
18	PARAMETER_MAX_RPM	Tachometer scale (default 8000).
19	PARAMETER_NORMAL_RESISTANCE_COOLANT	Coolant sensor resistance at 25 °C.
20	PARAMETER_NORMAL_RESISTANCE_OIL	Oil sensor resistance at 25 °C.
21	PARAMETER_NORMAL_RESISTANCE_AMB	Ambient sensor resistance at 25 °C.
23	PARAMETER_DOT_OFF	Clock colon behaviour (0 blink, 1 solid).
24	PARAMETER_BACKLIGHT_ON	Switch on backlight with low beam.
25	PARAMETER_M_D_FILTER	Median filter constant (legacy).
26	PARAMETER_COOLANT_MAX_R	Coolant “full-scale” temperature threshold.
27	PARAMETER_COOLANT_MIN_R	Coolant “1 bar” temperature threshold.
31–33	PARAMETER_MAINCOLOR_[RGB]	Interface colour components (Digifiz Next only).
37	PARAMETER_RPM_FILTER	RPM filtering aggressiveness.
128	PARAMETER_READ_ADDITION	Add to read any parameter.
255	PARAMETER_SET_HOUR	Set clock hours (24-hour).

Continued on next page

Table 10.4: Classic Digifiz Replica GEN1 configuration commands. (Continued)

254	PARAMETER_SET_MINUTE	Set clock minutes.
253	PARAMETER_RESET_DAILY_MILEAGE	Reset the trip odometer.
252	PARAMETER_RESET_DIGITAL	Factory reset and memory initialisation.

The Serial Bluetooth Terminal quick buttons are convenient for routine actions such as toggling automatic brightness (13 0 and 13 1) or writing colour values. Keep values above 60 % brightness only for short tests to preserve LED life.

Chapter 11

Typical situations for setting up the Digifiz Replica GEN1

Before troubleshooting, confirm that the dashboard is the classic Digifiz Replica GEN1 ([chapter 10](#)). Digifiz Replica Next (or Digifiz Next) panels use a Wi-Fi portal and are covered in [chapter 9](#).

Bluetooth module not detected Pair with the dashboard's Bluetooth Classic interface (it normally advertises as Digifiz). Serial Bluetooth Terminal for Android remains the recommended tool: configure the end-of-line character as LF and avoid BLE-only scanners, which cannot discover the module.

iPhone or iPad cannot connect Replica GEN1 dashboards use Bluetooth 2.0 and are incompatible with iOS devices. Use an Android phone or a computer running a Bluetooth serial utility.

Commands ignored on 2024+ firmware Unlock the command parser by sending 234 123, then repeat the desired sequence. Store quick-access buttons in Serial Bluetooth Terminal for the values you adjust frequently.

Speed reading too high or low Connect through Serial Bluetooth Terminal, drive at an indicated 100 km/h, and note the GPS speed. Send 1 <gps_value> (for example, 1 85) so `PARAMETER_SPEEDCOEFFICIENT` matches the verified GPS speed.

RPM reading incorrect Firmware prior to 2024 expects 0 <value> while current releases use 22 <value>. Audi engines typically need 22 3000; halve or double the value (for example, 22 1500 or 22 6000) until the display matches the tachometer.

Increase brightness Disable automatic control with 13 0 and raise the manual level with 14 <value>. Values between 45 and 55 brighten the display substantially; avoid levels above 60 to preserve LED life. Re-enable the photodiode later with 13 1.

Setting the clock Use Serial Bluetooth Terminal to send 255 <hours> followed by 254 <minutes>. Examples: 255 23, 254 55 sets 23:55; 255 14, 254 30 sets 14:30; 255 2, 254 28 sets 02:28.

Fuel gauge issues Disconnect the vehicle battery before probing.

- If the display drifts from 60 to 0, measure the sender resistance between the harness pin and ground; valid readings are typically 30–300 Ω . Clean the connector and confirm the signal reaches the main board.
- If the gauge is pegged full, look for a short to ground below 5 Ω on the sender line and repair it.
- If the reading never changes, compare the sender resistance with full and empty tanks. Replace the sensor if it stays constant.

Fuel flow values seem wrong The flow channel is emulated unless an intake-manifold pressure sensor is fitted. Treat the reading as indicative rather than absolute.

Coolant gauge inaccurate Adjust `PARAMETER_COOLANT_MIN_R` and `PARAMETER_COOLANT_MAX_R`. Example: 27 30 shortens the scale so that the “1 bar” mark aligns with roughly 30 °C.

Oil or ambient temperature readings missing A reading of -999 or a stuck value indicates a sensor issue. With the battery disconnected and the engine cold, measure the sensor resistance between the harness pin and ground. Oil sensors should read about 2 k Ω $\pm$ 0.3 k Ω ; ambient sensors about 10 k Ω $\pm$ 2 k Ω . Adjust `PARAMETER_NORMAL_RESISTANCE_OIL` (command 20) or `PARAMETER_NORMAL_RESISTANCE_AMB` (command 21) if the display needs fine-tuning. Persistent faults should be documented with `adc 0` logs and escalated to PHOL-LABS Kft support.

Collect raw sensor data with `adc 0` if the problem persists and share the results with the dashboard developers for analysis.

Chapter 12

Marking and sealing

- Each dashboard may be marked with the model number corresponding to its instrument-cluster variant.
- Export batches can include additional markings applied by the importing organisation.
- The Digifiz Replica GEN1 is not sealed; no tamper seals are fitted at the factory.

Chapter 13

Package

1. For transport, wrap the instrument panel set in bubble wrap and place it in a rigid cardboard box.
2. Alternative packaging is acceptable provided it protects the dashboard during transport and storage.

Chapter 14

Storage and transportation rules

- Transportation conditions must comply with the general freight rules applicable to each transport mode (GOST 23216-78).
- Packaged dashboards may be shipped by road, rail, river, or air transport.
- Store the instrument panel inside the vehicle cabin or in a heated room between 15 °C and 40 °C. Protect the unit from direct sunlight, although storage behind vehicle glass is permissible.

Appendix A

Reference Tables

A.1 Classic Digifiz Replica GEN1 Command Reference

The classic Replica firmware shares most commands with Digifiz Next. Commands 31–33 (colour control) are only active on Digifiz Next units; other commands apply equally to both generations.

A.2 Classic Replica GEN1 Default Values

Command	Name	Description
22(or0)	PARAMETER_RPMCOEFFICIENT	Engine RPM calibration factor.
1	PARAMETER_SPEEDCOEFFICIENT	Speed calibration factor.
2	PARAMETER_COOLANTTHERMISTORB	Coolant thermistor beta coefficient.
3	PARAMETER_OILTHERMISTORB	Oil thermistor beta coefficient.
4	PARAMETER_AIRTHERMISTORB	Ambient thermistor beta coefficient.
5	PARAMETER_TANKMINRESISTANCE	Minimum fuel sender resistance.
6	PARAMETER_TANKMAXRESISTANCE	Maximum fuel sender resistance.
7--10	PARAMETER_TAU_X	Filtering constants for coolant, oil, air, and fuel level.
11	PARAMETER_MILEAGE	Total odometer value.
12	PARAMETER_DAILY_MILEAGE	Trip odometer.
13	PARAMETER_AUTO_BRIGHTNESS	Automatic brightness enable.
14	PARAMETER_BRIGHTNESS_LEVEL	Manual brightness level.
15	PARAMETER_TANK_CAPACITY	Fuel tank capacity.
16	PARAMETER_MFA_STATE	Active MFA mode.
17	PARAMETER_BUZZER_OFF	Disable buzzer (Replica only).
18	PARAMETER_MAX_RPM	Tachometer scaling (7000 default).
19--21	PARAMETER_NORMAL_RESISTANCE_X	Sensor resistances at 25 °C for coolant, oil, and ambient inputs.
23	PARAMETER_DOT_OFF	Clock colon behaviour.
24	PARAMETER_BACKLIGHT_ON	Enable backlight when low beam is active.
25	PARAMETER_M_D_FILTER	Median filter constant.
26	PARAMETER_COOLANT_MAX_R	Coolant sensor threshold for full-scale indication.
27	PARAMETER_COOLANT_MIN_R	Coolant sensor threshold for “1 bar” indication.
31--33	PARAMETER_MAINCOLOR_[RGB]	UI colour components (Digifiz Next only).
37	PARAMETER_RPM_FILTER	RPM filtering aggressiveness.
128	PARAMETER_READ_ADDITION	Add 128 to read the current value of a command.
255	PARAMETER_SET_HOUR	Set clock hours.
254	PARAMETER_SET_MINUTE	Set clock minutes.
253	PARAMETER_RESET_DAILY_MILEAGE	Reset trip odometer.
252	PARAMETER_RESET_DIGITAL	Factory reset of stored parameters.

Table A.1: Primary configuration commands for classic Digifiz Replica GEN1 dashboards.

Parameter	Default	Notes
PARAMETER_RPMC OEFFICIENT	3000	
PARAMETER_SPEE DCOEFFICIENT	100	
PARAMETER_COOL ANTTHERMISTORB	4000	
PARAMETER_OILTH ERMISTORB	4000	
PARAMETER_AIRT HERMISTORB	3812	3600 on Gen 2.
PARAMETER_TANK MINRESISTANCE	35	Ohms.
PARAMETER_TANK MAXRESISTANCE	265	Ohms.
PARAMETER_TAU_ COOLANT	2	
PARAMETER_TAU_ OIL	2	
PARAMETER_TAU_ AIR	2	
PARAMETER_TAU_ TANK	2	
PARAMETER_MILE AGE	Vehicle-specific	Preserve existing odometer.
PARAMETER_DAILY _MILEAGE	0	
PARAMETER_AUTO _BRIGHTNESS	1	Enabled.
PARAMETER_BRIG HTNESS_LEVEL	7 or 13	Typical values for Gen 1/1.5.
PARAMETER_TANK _CAPACITY	63	Litres.
PARAMETER_MFA_ STATE	0	
PARAMETER_BUZZ ER_OFF	1	Buzzer disabled.
PARAMETER_MAX_ RPM	8000	7000 on earlier clusters.
PARAMETER_NORM AL_RESISTANCE_C OOLANT	1000	Ω at 25 °C.
PARAMETER_NORM AL_RESISTANCE_OI L	1000	Ω at 25 °C.
PARAMETER_NORM AL_RESISTANCE_A MB	2991	Ω at 25 °C.
PARAMETER_DOT_ OFF	0	Blinking colon.
PARAMETER_BACK LIGHT_ON	1	Enabled.
PARAMETER_M_D_ FILTER	65535	
PARAMETER_COOL ANT_MAX_R	120	°C.
PARAMETER_COOL ANT_MIN_R	60	°C.
PARAMETER_MAIN COLOR_[RGB]	–	Colour commands not active on classic Replica.
PARAMETER_RPM_ FILTER	70	
PARAMETER_UPTI ME	0	

Appendix B

Digifiz Next Parameter Catalogue

All parameters in this catalogue are located on the Settings tab. Load a parameter before editing it, and load and edit only one parameter at a time. The identifiers below are firmware settings; similarly named parameter commands in the preceding reference are separate interfaces and must not be treated as the same item. UI labels and firmware descriptions are reproduced verbatim so that they can be matched to the device regardless of the manual language.

Table B.1: Digifiz Next firmware parameters in the Settings tab

Identifier	UI label	Description	Type / default	Limits / notes
option_mfa_manufacturer	MFA Option Enable	Enable MFA steering columns switch	U8 / 1	0–1
option_miles	Miles option	Enable Miles option	U8 / 0	0–1
option_gallons	Gallons option	Enable Gallons option	U8 / 0	0–1
option_fahrenheit	Fahrenheit option	Enable Fahrenheit degrees option	U8 / 0	0–1
option_kelvin	Kelvin option	Enable Kelvin degrees option	U8 / 0	0–1
option_lbar	–	Reserved firmware option	U8 / 0	0–1
option_linear_fuel	Linear fuel calculation	Calculate the fuel level by simple linear formula	U8 / 0	0–1
option_testmode_on	Test mode	Enable testmode	U8 / 0	0–1
option_invert_oil03_input	Invert 0.3 bar input	Invert the 0.3 bar oil pressure input (1 is treated as 0 and 0 as 1)	U8 / 0	0–1

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Table B.1: Digifiz Next firmware parameters in the Settings tab (Continued)

Identifier	UI label	Description	Type / default	Limits / notes
rpmOptions_redline_segments	Redline segments	Redline segments number	U8 / 8	0–31
rpmOptions_diesel_line	Diesel option	Diesel-style RPM gauge	U8 / 0	0–1
rpmOptions_7k_line	7k option	7k-style RPM gauge	U8 / 0	0–1
tempOptions_red_segments	Red segments count	Temperature red segments count	U8 / 0	0–3
tempOptions_sensor_connected_ind	Indicate temp. sensor	Display that sensor is connected by indicating 1 segment	U8 / 0	0–1
tempOptions_oil_atten	Lower atten. enable	Enable lower attenuator for Oil channel	U8 / 0	0–1
tempOptions_alarm_function	Temp. alarm function	Blink the temperature gauge if alarm	U8 / 0	0–1
checkEngineAction	Check engine action	0 = no action; 1 = blink temperature indicator (1 s); 2 = blink speedometer red (1 s)	U8 / 0	0–2
signalOptions_use_blink_alt_in	Other blink	Use alternative separate blinker inputs	U8 / 0	0–1
signalOptions_enable_firmware_blinkers	Firmware blinkers	Blink turn indicators in firmware (0.5 s period)	U8 / 0	0–1
signalOptions_enable_touch_sensor	Touch sensor option	Enable wireless touch sensor	U8 / 0	0–1
signalOptions_invert_light_input	Invert light signal input	Invert behaviour of brightness adjustment	U8 / 0	0–1
signalOptions_enable_consumption_sensor	Consumption sensor option	Enable external consumption MAF sensor	U8 / 0	0–1
signalOptions_indicator_filter_cycles	Indicator filter cycles	Cycles required to confirm general indicator changes	U16 / 0	0–65535

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Table B.1: Digifiz Next firmware parameters in the Settings tab (Continued)

Identifier	UI label	Description	Type / default	Limits / notes
signalOptions_turn_filter_cycles	Turn signal filter cycles	Cycles required to confirm turn indicator changes	U16 / 0	0–65535
rpmQuadraticCoefficient	RPM Quadratic coefficient	Quadratic term coefficient for RPM calibration (divide by 100000)	U16 / 0	0–65535
rpmCoefficient	RPM Coefficient	Coefficient for RPM calibration	U16 / 1500	0–65535
speedCoefficient	Speed coefficient	Coefficient for speed calibration	U16 / 100	0–65535
coolantThermistorB	Coolant thermistor beta	Beta-coefficient for coolant temperature equation	U16 / build value	Firmware constant
oilThermistorB	Oil thermistor beta	Beta-coefficient for oil temperature equation	U16 / build value	Firmware constant
airThermistorB	Air thermistor beta	Beta-coefficient for outdoor temperature equation	U16 / build value	Firmware constant
tankMinResistance	Fuel tank min. resistance	Minimum resistance of fuel sensor	U16 / 35	Ohms
tankMaxResistance	Fuel tank max. resistance	Maximum resistance of fuel sensor	U16 / 265	Ohms
tauCoolant	Coolant LPF coeff.	Filter coefficient for coolant temperature	U16 / 2	0–65535
tauOil	Oil LPF coeff.	Filter coefficient for oil temperature	U16 / 2	0–65535
tauAir	Air LPF coeff.	Filter coefficient for outdoor temperature	U16 / 2	0–65535
tauTank	Tank LPF coeff.	Filter coefficient for fuel level	U16 / 2	0–65535
maxRPM	Maximum RPM	Maximum RPM (8000, 7000, or 6000)	U16 / 8000	Listed values
coolantMin	Min. coolant temp.	Minimum coolant temperature (1 segment)	U16 / 60	Degrees

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Table B.1: Digifiz Next firmware parameters in the Settings tab (Continued)

Identifier	UI label	Description	Type / default	Limits / notes
coolantMax	Max. coolant temp.	Maximum coolant temperature (maximum segments)	U16 / 120	Degrees
medianDispFilterThreshold	N/a	Not used	U16 / 65535	Reserved
autoBrightness	Auto brightness	Enable auto brightness calculation	U8 / 1	0–255
brightnessLevel	Brightness level	Manual brightness level	U8 / 25	0–70
brightnessSignalMin	Light signal min	ADC value for minimum brightness	U16 / 300	ADC value
brightnessSignalMax	Light signal max	ADC value for maximum brightness	U16 / 800	ADC value
brightnessMin	Brightness min	Minimum auto brightness level	U8 / 6	0–255
brightnessMax	Brightness max	Maximum auto brightness level	U8 / 25	0–255
brightnessSpeed	Brightness speed	Regulation speed (percent)	U8 / 10	0–100
tankCapacity	Fuel tank capacity	Fuel tank capacity in liters	U8 / 63	Litres
fuel_low_threshold_liters	Low fuel threshold (L)	Fuel level in liters that triggers refuel indicator	U8 / 10	Litres
fuel_critical_threshold_liters	Critical fuel threshold (L)	Fuel level in liters that triggers refuel blinking	U8 / 5	Litres
fuel_low_threshold_gallons	Low fuel threshold (gal)	Fuel level in gallons that triggers refuel indicator	U8 / 3	Gallons
fuel_critical_threshold_gallons	Critical fuel threshold (gal)	Fuel level in gallons that triggers refuel blinking	U8 / 2	Gallons
mfa_sensor	MFA sensor type	0 = consumption; 1 = barometer; 2 = wideband lambda; 3 = fuel pressure	U8 / 0	0–5
mfa_clock_sensor	MFA clock sensor type	0 = clock; 1 = barometer; 2 = wideband lambda; 3 = fuel pressure	U8 / 0	0–5

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Table B.1: Digifiz Next firmware parameters in the Settings tab (Continued)

Identifier	UI label	Description	Type / default	Limits / notes
Vblow	Barometer V low	Low voltage calibration point for barometer	float / 0.25	0.0–3.3 V
Vbhigh	Barometer V high	High voltage calibration point for barometer	float / 2.25	0.0–3.3 V
Blow	Barometer low	Pressure at low voltage point	float / 0.0	–20.0–20.0
Bhigh	Barometer high	Pressure at high voltage point	float / 8.0	–20.0–20.0
Vfplow	Fuel pressure V low	Low voltage calibration point for fuel pressure	float / 0.0	0.0–3.3 V
Vfphigh	Fuel pressure V high	High voltage calibration point for fuel pressure	float / 3.3	0.0–3.3 V
FPlow	Fuel pressure low	Pressure at low voltage point	float / 0.0	–20.0–20.0
FPhigh	Fuel pressure high	Pressure at high voltage point	float / 10.0	–20.0–20.0
Viplow	Intake pressure V low	Low voltage calibration point for intake pressure	float / 0.0	0.0–3.3 V
Viphigh	Intake pressure V high	High voltage calibration point for intake pressure	float / 3.3	0.0–3.3 V
IPlow	Intake pressure low	Pressure at low voltage point	float / 84749.0	–20000– 250000
IPhigh	Intake pressure high	Pressure at high voltage point	float / 18247.4	–20000– 250000
Vllow	Lambda V low	Low voltage calibration point for wideband lambda	float / 0.0	0.0–3.3 V
Vlhigh	Lambda V high	High voltage calibration point for wideband lambda	float / 4.5	0.0–5.0 V
AFRlow	Lambda AFR low	AFR at low voltage point	float / 10.0	0.0–40.0
AFRhigh	Lambda AFR high	AFR at high voltage point	float / 20.0	0.0–40.0

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Table B.1: Digifiz Next firmware parameters in the Settings tab (Continued)

Identifier	UI label	Description	Type / default	Limits / notes
mfaState	MFA function state	Selected MFA function	U8 / –	Runtime value
buzzerOff	Buzzer option	Enable or disable embedded buzzer	U8 / 1	0–255
mfaBlock	MFA Block ind.	Selected MFA memory block	U8 / –	Runtime value
displayDot	Blinking dot	Display blinking dot	U8 / –	Runtime value
backlight_on	Bkclt power	Enable backlight power	U8 / 1	0–255
speedColorChangeEnable	Speed color alert	Enable speed color change above threshold	U8 / 0	0–1
speedColorChangeThreshold	Speed alert threshold	Speed value that enables alert color	U16 / 120	0–400
speedAlertColor_r	Speed alert R	Speed alert color red component	U8 / 255	0–255
speedAlertColor_g	Speed alert G	Speed alert color green component	U8 / 0	0–255
speedAlertColor_b	Speed alert B	Speed alert color blue component	U8 / 0	0–255
mainc_r	Main R color	Main theme red color component	U8 / 180	0–255
mainc_g	Main G color	Main theme green color component	U8 / 240	0–255
mainc_b	Main B color	Main theme blue color component	U8 / 6	0–255
backc_r	Back R color	Main theme backlight red color component	U8 / 180	0–255
backc_g	Back G color	Main theme backlight green color component	U8 / 240	0–255
backc_b	Back B color	Main theme backlight blue color component	U8 / 6	0–255
ledEffect_type	LED effect	LED effect type	U8 / 0	0–255
ledEffect_hue	Effect hue	Base hue value	U8 / 0	0–255
ledEffect_saturation	Effect saturation	Base saturation	U8 / 255	0–255
ledEffect_value	Effect value	Base brightness	U8 / 30	0–255

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Table B.1: Digifiz Next firmware parameters in the Settings tab (Continued)

Identifier	UI label	Description	Type / default	Limits / notes
coolantThermistorDefRes	Coolant default resistance	Default coolant sensor resistance at 25 degrees C	U16 / build value	Ohms
oilThermistorDefRes	Oil default resistance	Default oil temperature sensor resistance at 25 degrees C	U16 / build value	Ohms
ambThermistorDefRes	Air default resistance	Default outdoor temperature sensor resistance at 25 degrees C	U16 / build value	Ohms
rpmFilterK	RPM LPF coeff.	LPF coefficient for RPM calculation	U16 / 70	0–65535
rpmFallingFilterK	RPM LPF coeff. fall	LPF coefficient used when RPM decreases	U16 / 200	0–65535
speedFilterK	LPF coefficient for speed calculation	LPF coefficient for speed calculation	U16 / –	Build-dependent
speedDisplayUpdateDivider	Speed display update divider	Number of 32 Hz main-loop cycles between displayed speed updates	U16 / 16	1–320
coolantThermistorPullUpRes	Coolant pull-up	Pull-up resistance for coolant temperature sensor	U16 / 220	Ohms
oilThermistorPullUpRes	Oil pull-up	Pull-up resistance for oil temperature sensor	U16 / 220	Ohms
oilThermistorSeriesRes	Oil series	Series resistance between oil sensor and ground	U16 / 0	Ohms
ambThermistorPullUpRes	Amb. air pull-up	Pull-up resistance for ambient temperature sensor	U16 / 3300	Ohms
useCustomScheme	Custom scheme	Enable custom display scheme	U8 / 0	0–1
wifi_auto_off	WiFi Auto Off	Turn off WiFi after 1 minute of ignition ON	U8 / 0	0–1
rpm_algorithm	RPM algorithm	0 = legacy/unstable; 1 = stable input; 2 = glitch-filtered input	U8 / 0	0–2

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Table B.1: Digifiz Next firmware parameters in the Settings tab (Continued)

Identifier	UI label	Description	Type / default	Limits / notes
rpm_sanity_check	RPM sanity check	Maximum valid RPM value used as sanity threshold	U16 / 3000	0–65535
rpm_debounce_ticks	RPM debounce ticks	Debounce threshold in timer ticks for RPM edge filtering	U16 / 50	Timer ticks
rpm_min_low_us	RPM min low us	Minimum low pulse width for glitch-filtered RPM input	U16 / 100	Microseconds
rpm_min_high_us	RPM min high us	Minimum high pulse width for glitch-filtered RPM input	U16 / 100	Microseconds
r_gear_coefficient	Rear gear coefficient	Proportion between speed and gear	U32 / 9999999	Default: N/A
gear1_coefficient	1-st gear coefficient	Proportion between speed and gear	U32 / 1000	Default: 020
gear2_coefficient	2-nd gear coefficient	Proportion between speed and gear	U32 / 2000	Default: 020
gear3_coefficient	3-rd gear coefficient	Proportion between speed and gear	U32 / 3000	Default: 020
gear4_coefficient	4-th gear coefficient	Proportion between speed and gear	U32 / 4000	Default: 020
gear5_coefficient	5-th gear coefficient	Proportion between speed and gear	U32 / 5000	Default: 020
gear6_coefficient	6-th gear coefficient	Proportion between speed and gear	U32 / 10000	Default: no gear
speed_rpm_mode	Speed source mode	0 = speed input; 1 = RPM with 3-rd gear coefficient; 2 = dynamic RPM/gear coefficients	U8 / 0	0–2; reboot
option_gear_change_indicator	Gear change indicator	Blink gear on speedometer when gear changes	U8 / 0	0–1
option_gear_indicator_in_refuel	Gear indicator in refuel	Show gear indicator in refuel segment instead of speed display	U8 / 0	0–1

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Table B.1: Digifiz Next firmware parameters in the Settings tab (Continued)

Identifier	UI label	Description	Type / default	Limits / notes
option_refuel_blink	Blink refuel indicator	Blink refuel (R) indicator when fuel is low	U8 / 0	0–1
speed_meas_period	Speedometer measurement period in ms	Measurement period for PCNT	U32 / 1000	250–2000; reboot
speed_meas_window	Speedometer measurement window in ms	Measurement window for PCNT	U32 / 1000	250–2000; reboot
stable_speed_input	Stable Speed input	Enable for electronic speed signal	U8 / 0	0–1
filtered_speed_input	Filtered Speed input	Enable speed signal filtering (disable stable input)	U8 / 1	0–1
speed_min_threshold_value	Minimum speed threshold	Minimum speed shown as non-zero	U16 / 5	0–400

Appendix C

Document Change Log

Change	Affected sheets	Date
1	04.10.2022	04 Oct 2022
2	31.08.2023	31 Aug 2023
3	05.08.2024	05 Aug 2024
4	LaTeX document introduced.	22.09.2025

Table C.1: Document change registration sheet.